Radioactive Decay and Half-Life

Nuclear equations, radiation and half-life

Name _____ Class _____ Date _____

RULES AND DATA

Alpha $^{4}_{2}\mathrm{He}$: A − 4, Z − 2 · Beta $^{0}_{-1}\mathrm{e}$: Z + 1 · Positron $^{0}_{+1}\mathrm{e}$: Z − 1 · Gamma: no change

$N = N_0 \times (½)^{n}$, $n = t \div t_{½}$ · $t = t_{½} \times \ln(N_0/N) \div \ln 2$

Z: N 7 · O 8 · Ca 20 · Ni 28 · Xe 54 · Rn 86 · Th 90 · Pa 91 · U 92

Use the same time unit for t and $t_{½}$. Show units on every quantity.

A. Understanding

1. Define (a) radioactivity (b) half-life. (c) Why can we predict the half-life of a large sample but not the moment when one particular nucleus will decay?

B. Nuclei and nuclear equations

2. Give the numbers of protons and neutrons in: (a) $^{235}_{92}\mathrm{U}$ (b) $^{60}_{27}\mathrm{Co}$ (c) $^{131}_{53}\mathrm{I}$ (d) $^{222}_{86}\mathrm{Rn}$

3. Complete each equation: (a) $^{226}_{88}\mathrm{Ra} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$ (b) $^{131}_{53}\mathrm{I} \longrightarrow ? + {}^{0}_{-1}\mathrm{e}$ (c) $^{18}_{9}\mathrm{F} \longrightarrow ? + {}^{0}_{+1}\mathrm{e}$ (d) $^{60}_{27}\mathrm{Co} \longrightarrow ? + {}^{0}_{-1}\mathrm{e}$ (e) $^{239}_{94}\mathrm{Pu} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$ (f) $^{40}_{19}\mathrm{K} \longrightarrow ? + {}^{0}_{-1}\mathrm{e}$

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4. Identify X: (a) $^{234}_{90}\mathrm{Th} \longrightarrow {}^{234}_{91}\mathrm{Pa} + \mathrm{X}$ (b) $^{210}_{84}\mathrm{Po} \longrightarrow {}^{206}_{82}\mathrm{Pb} + \mathrm{X}$ (c) $^{11}_{6}\mathrm{C} \longrightarrow {}^{11}_{5}\mathrm{B} + \mathrm{X}$

5. Uranium-238 ($^{238}_{92}\mathrm{U}$) decays by alpha emission, then beta, then beta again. Write the three equations and identify the final nucleus.

C. Types of radiation

6. Compare alpha, beta and gamma radiation: what each one is, its charge, and what stops it. Explain why an alpha emitter is most dangerous inside the body.

D. Half-life

7. A patient is given technetium-99m with an activity of 800 MBq ($t_{1/2}$ = 6.01 h). What is its activity after 24.04 h?

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8. (a) The activity of a sample falls from 1600 Bq to 400 Bq in 30.0 min. Find its half-life. (b) Radon-222 has $t_{1/2}$ = 3.82 d. What percentage of a sample remains after 10.0 d?

E. Challenge

9. A sample of carbon from an ancient wooden bowl has a carbon-14 activity of 0.0850 Bq per gram of carbon. Carbon from fresh wood measured in the same laboratory gives 0.230 Bq per gram. Estimate the age of the bowl ($t_{1/2}$ of carbon-14 = 5730 yr).

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